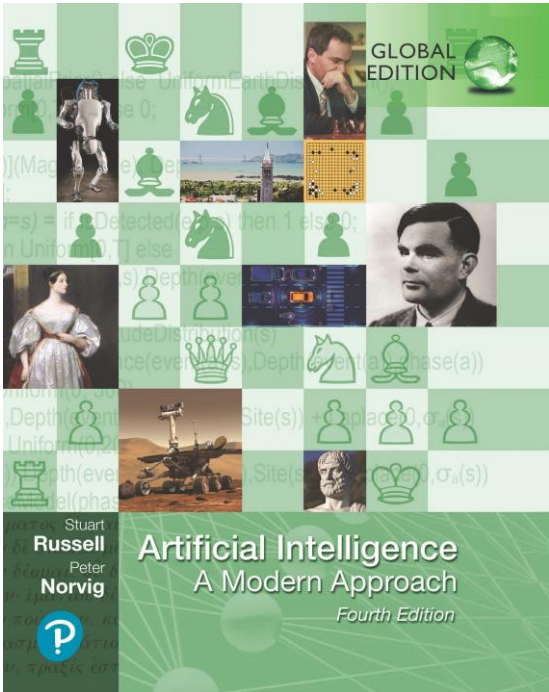


Chapter 3

Solving Problems by Searching

Homework # 2

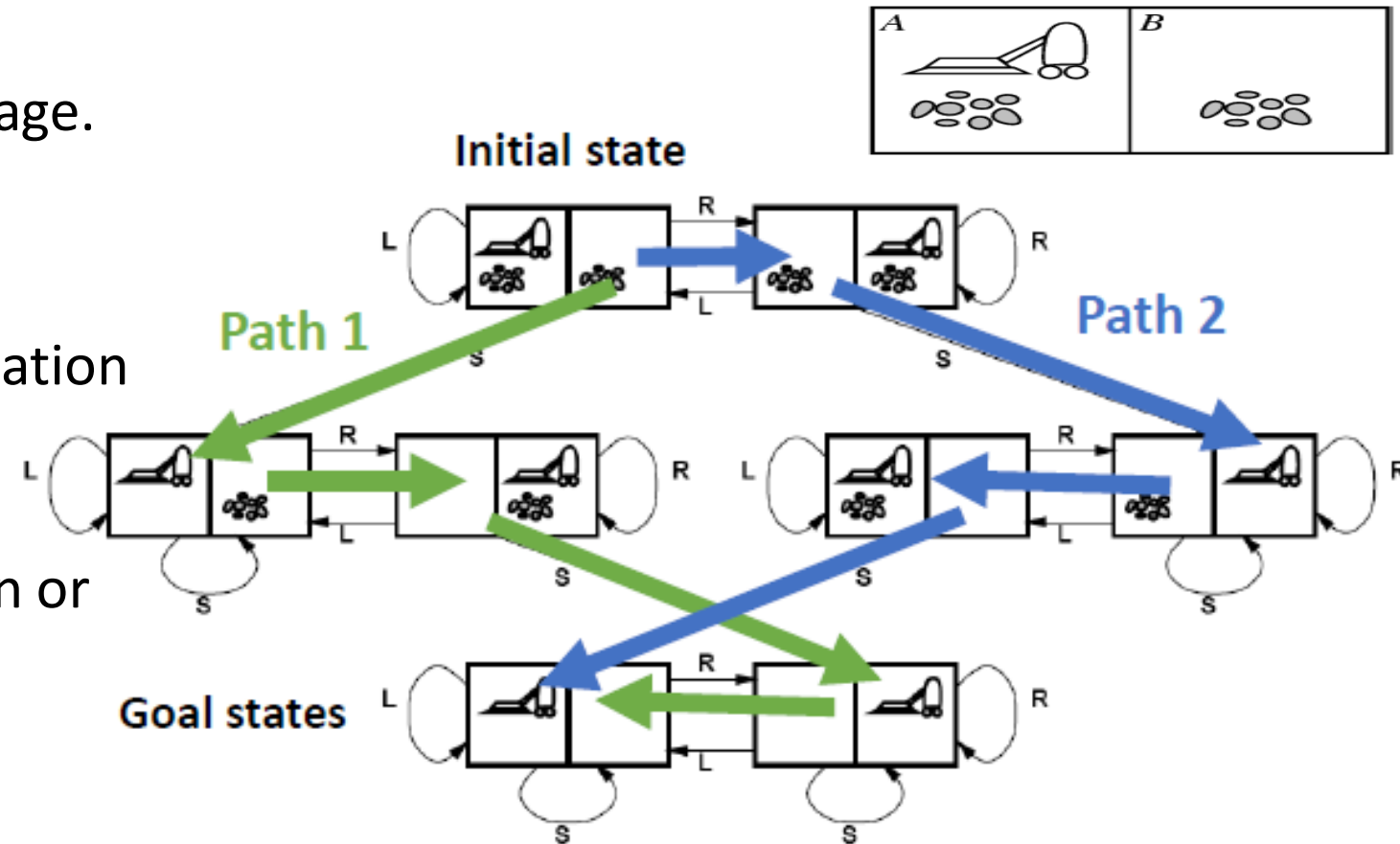


Implementation of the vacuum Cleaner Intelligent Agent

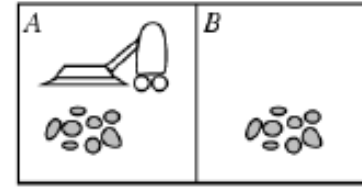
- Write a Python program to implement the vacuum cleaner Intelligent agent where **Breadth-first search** and **Uniform-cost search** are defined using Python.
 - Program should be flexible to select any state as initial state, find the lowest cost path to reach the final state.
 - Use the definitions on the next page.

- Important definitions

- **Initial State:** Defined by agent location and dirt location
- **Actions:** Left, right, suck
- **Transition model:** Clean a location or move
- **Goal state:** All locations are clean
- **Path cost:** E.g., number of actions



Tree search



- State Space Size

- Examples: What is the state space size?
- Dirt
 - Permutation: A and B are different rooms, order does matter!
 - With repetition: Dirt can be in both rooms
 - There are 2 options (clean/dirty) $\rightarrow n = 2$
 - $r = 2 \rightarrow$ number of rooms
 - Total size = $n^r = 2^2 = 4$

- Robot location

- Can be in 1 out of 2 rooms $\rightarrow 2$
- Total: $2 \times 2^2 = 2^3 = 8$

